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Efficient frequency-decomposed transformer via large vision model guidance for surgical image desmoking

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ABSTRACT

Surgical image restoration plays a vital clinical role in improving visual quality during surgery, particularly in minimally invasive procedures where the operating field is frequently obscured by surgical smoke. However, surgical image desmoking still has limited progress in algorithm development and customized learning strategies. In this regard, this work focuses on the task of desmoking from both theoretical and practical perspectives. First, we analyze the intrinsic characteristics of surgical smoke degradation: (1) spatial localization and dynamics, (2) distinguishable frequency-domain patterns, and (3) the entangled representation of anatomical content and degradative artifacts. These observations motivated us to propose an efficient frequency-aware Transformer framework, namely SmoRestor, which aims to separate and restore true anatomical structures from complex degradations. Specifically, we introduce a high-order Fourier-embedded neighborhood attention transformer that enhances the model's ability to capture structured degradation patterns across both spatial and frequency domains. Besides, we utilize the semantic priors encoded by large vision models to disambiguate content from degradation through targeted guidance. Moreover, we propose an innovative transfer learning paradigm that injects knowledge from large models to the main network, enabling it to effectively distinguish meaningful content from ambiguous corruption. Experimental results on both public and in-house datasets demonstrate substantial improvements in quantitative performance and visual quality. The source code will be available.

1. Introduction

Surgical image restoration, though related to natural image enhancement, presents distinct challenges due to the specialized environment of endoscopic surgery. In addition to improving image quality, surgical restoration must recover fine-grained anatomical details, such as blood vessels and tissue textures, which are vital for intraoperative decision-making (Guo et al., 2025; Pei et al., 2024a). Among the various degradation types, *desmoking* (Sengar et al., 2021; Chen et al., 2019; Wang et al., 2023; Venkatesh et al., 2020; Pan et al., 2022; Xia et al., 2024a) remains one of the most prominent issues. Smoke generated by cauterization and tissue dissection severely impairs visibility, blending artifacts with anatomical content and making real-time restoration both critical and challenging (Ma et al., 2025; He et al., 2010; Wang et al., 2019; Salazar-Colores et al., 2020). Moreover, edge devices such as laparoscopic cameras often operate under stringent computational constraints, necessitating lightweight yet efficient algorithms that can function with minimal latency in surgical environments (Wu et al., 2023; Zheng et al., 2023; Chen et al., 2024a; Fang et al., 2024).

Unlike natural scenes with relatively simple degradations, surgical scenes exhibit complex and spatially heterogeneous distortions (Pei et al., 2024b; Guo et al., 2024b). As illustrated in Fig. 1, smoke image degradation often exhibits regional and dynamic characteristics, frequently obscuring important features such as blood vessels and lesions (Pan et al., 2022). These degraded regions often share visual similarity with anatomical textures, making it difficult to distinguish smoke from content (Wu et al., 2024). In real laparoscopic surgery, where accuracy and efficiency are paramount, the challenge lies in restoring visual clarity while preserving crucial spatial details. Although surgical restoration is important in clinical practice, desmoking has received limited attention in algorithm research (Sengar et al., 2021; Chen et al., 2019; Wang et al., 2023), particularly under constraints posed by real-world deployment (Venkatesh et al., 2020; Pan et al., 2022; Wu et al., 2024).

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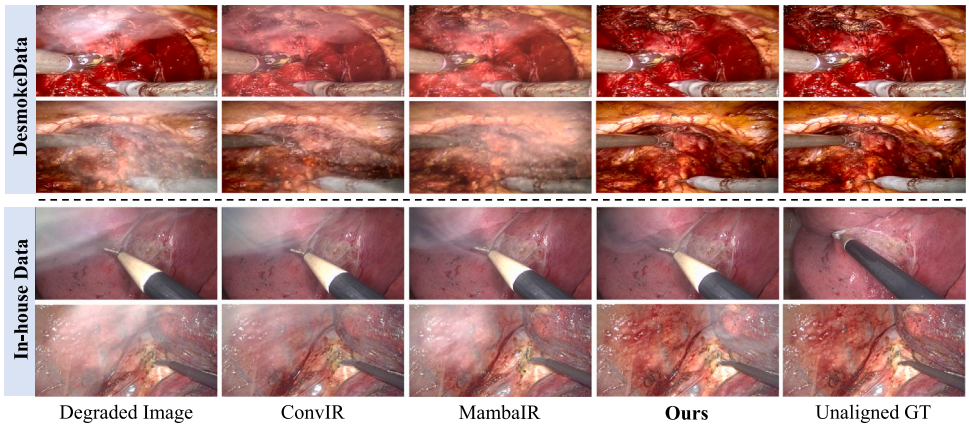

Fig. 1. Visualization of desmoking performance between representative image restoration models and SmoRestor in real surgical degradation scenarios. We can see that our method exhibits more noticeable restoration capabilities on both DesmokeData (Xia et al., 2024b) and in-house datasets.

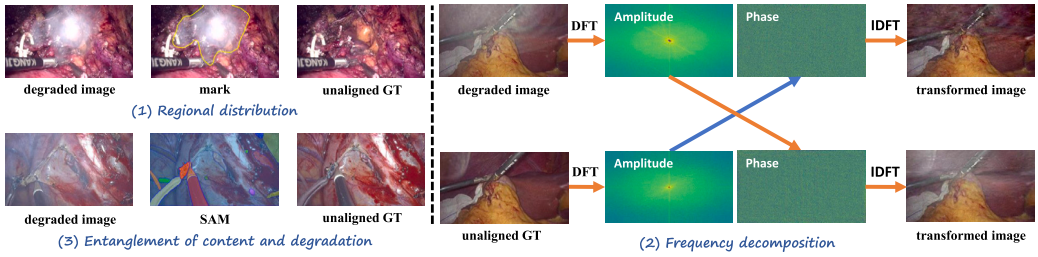

Fig. 2. Illustration of the inherent properties of surgical image desmoking, including (1) regional disruption, (2) frequency decomposition, and (3) entanglement of content and degradation. At the data level, there exists an unalignment between the degraded inputs and the ground truth. Zoom in for details.

As shown in Fig. 2, surgical smoke exhibits unique degradation properties: (1) *regional disruption*, where certain parts of the image are disproportionately degraded, (2) *frequency decomposition*, where different spatial frequencies are impacted at varying levels, and (3) *entanglement of content and degradation*, where the entanglement of smoke and underlying anatomical content add another layer of difficulty in feature separation (Yang et al., 2024a).

According to the revisiting of the intrinsic degradation characteristics of surgical desmoking, we propose SmoRestor, a novel solution based on the efficient frequency-decomposed transformer through large vision models (Dehghani et al., 2023; Fang et al., 2023), enabling more precise modeling of both spatial and frequency domain distortions. Moreover, our framework balances surgical system latency constraints with the strength of large pre-trained models. During training, selected features are masked and reconstructed via an auxiliary branch guided by a large vision model SAM (Kirillov et al., 2023). This branch is removed at inference, enabling high performance without additional runtime cost. By combining semantic guidance, domain-specific degradation modeling, and efficient attention mechanisms, SmoRestor achieves both accuracy and speed. Designed for clinical deployment, it ensures visual clarity, fast execution, and practical reliability in real-world surgical scenarios. Beyond visual enhancement, the restored outputs can be seamlessly integrated into surgical workflows and downstream tasks. In particular, smoke-free frames provide clearer visualization for surgeons during decision-making and serve as reliable inputs for computer-assisted systems such as workflow recognition, bleeding detection, and anatomical localization. The key contributions of this work are fourfold:

- **Dedicated desmoking framework:** We propose a novel surgical restoration framework tailored for real-time desmoking. It addresses domain-specific degradation, ensuring computationally efficient deployment suitable for clinical practice.
- **Frequency-decomposed attention:** A frequency-decomposed transformer is designed that incorporates high-order Fourier-embedded features within a neighborhood attention to capture spatial-frequency interactions and disentangle smoke from content.
- **Lightweight transfering knowledge via large vision model:** We introduce a novel dual-branch training strategy that transfers semantic priors from large vision models into compact ones, enabling efficient and accurate desmoking for edge deployment.
- **Theoretically grounded design:** The proposed framework embraces principles from Fourier analysis and attention theory, providing theoretical guarantees on its stability and robustness in real-world surgical environments.

2. Related works

2.1. Natural image restoration

Natural image restoration technology has made great progress in addressing various image degradation problems such as noise, blur, and low resolution. Early works (Cui et al., 2024; Cai et al., 2016; Liu et al., 2019; Zhang and Patel, 2018; Ren et al., 2016) adopt convolutional neural networks (CNNs) to enhance image restoration. These network architectures emphasize feature adaptation and hierarchical feature extraction, which are critical for improving restoration performance. Subsequently, Cui et al. (2023) embedded a dual-domain selection mechanism based on the U-shaped CNN backbone to make full use of the discrepancies between degraded image pairs in both spatial and spectral domains. This method proved highly effective in practical scenarios where image quality is critical. As image degradation becomes more complex, various brand-new models have emerged, *e.g.*, Fourmer (Zhou et al., 2023), focusing on global modeling. The Fourier-domain framework integrates spatial and frequency domain information, demonstrating effectiveness for large-scale restoration, especially

for high-resolution images. Besides, Restormer (Zamir et al., 2022) exploits the transformer ability (Pei et al., 2022; Zhou et al., 2025b) to capture long-range dependencies for image restoration, which is a key factor in recovering fine details in high-resolution images. Despite these advances, many state-of-the-art methods are not optimized for resource-constrained environments. To alleviate this limitation, Guo et al. (2024a) proposed a Mamba-based (Gu and Dao, 2023) state-space model that provides both theoretical guarantees and practical efficiency even when dealing with large datasets. Sun et al. (2025) adopted a unique approach based on a histogram transformer that integrates spatial and frequency-domain information to address weather-induced degradation for image restoration in adverse weather conditions. Additionally, Choi et al. (2024) focused on lightweight restoration methods using transformer-based models, balancing computational cost and performance to meet the needs of real-time mobile applications. Furthermore, recent research in remote sensing has explored specialized architectures for challenging degradations. For instance, an encoder-free multiaxis physics-aware fusion network was proposed to tackle dehazing in remote sensing images (Wen et al., 2023a), a cross-level interaction and intra-level fusion network was introduced for remote sensing image dehazing (Wen et al., 2025), and an adaptive degradation-aware self-prompting model was developed for all-in-one weather-degraded image restoration (Wen et al., 2025b). These works highlight the growing interest in robust architectures that generalize across different degradation types. Recently, diffusion probabilistic models have also emerged as a powerful class of generative models for image restoration. By progressively reversing a noise corruption process, they produce highly realistic outputs. Representative works such as SR3 (Saharia et al., 2023), Palette (Saharia et al., 2022), and DiffBIR (Lin et al., 2024) demonstrate strong performance across super-resolution, inpainting, and blind restoration tasks.

Despite these advancements in natural image restoration, surgical image restoration still faces unique challenges. Surgical desmoking typically exhibits more complex degradations, which require specialized algorithms for content restoration and degradation removal. Moreover, surgical applications demand real-time processing and high precision, necessitating lightweight models capable of handling these complex degradations while ensuring efficient operation on resource-constrained edge devices.

2.2. Image restoration via foundation models

Foundation models have exhibited great potential in enhancing image restoration by providing contextual knowledge and high-level representations that complement pixel-level restoration. This integration has notably enhanced restoration quality, especially when dealing with unknown or complex degradation types. Potlapalli et al. (2023) first proposed the use of prompt-based learning for image restoration. The architecture utilizes specific prompts to encode the degraded information, which is then used to dynamically guide the restoration network to generalize to different types and levels of degradation. Building on this, MPerceiver (Ai et al., 2024) is proposed, which combines text prompts for holistic representation and visual prompts for multi-scale detail representation with the language-vision foundation model *i.e.*, CLIP (Radford et al., 2021), to adaptively respond to unknown degradations. Inspired by the integrated learning in foundation models, Li et al. (2024) contributed a million-scale image restoration dataset and introduced FoundIR to construct an integrated learning framework by combining a generalist model with multiple expert models, thereby taking another step towards foundation models. In the medical imaging domain, Yang et al. (2024a) presented an all-in-one medical image restoration network that incorporates an adaptive routing strategy to significantly reduce interference between tasks, achieving superior performance on tasks such as MRI super-resolution and CT denoising. Besides, A novel region-attention transformer (RAT) (Yang et al., 2024b) is introduced for various medical image restoration tasks. This approach utilizes a region-based multi-head self-attention combined with a large vision model to guide the model to adaptively focus on restoring difficult regions.

Although the success of foundation model-assisted restoration in natural and medical images, applying these models for surgical image restoration presents unique challenges. In surgical scenes, the need for lightweight operation and real-time processing is paramount, particularly for edge devices like laparoscopic cameras and endoscopes with limited computational resources. Harnessing foundational models to meet the requirements of real-time, efficient operations in surgical environments remains an open challenge.

2.3. High-order modeling and Fourier prior

In recent years, significant progress has been made in incorporating high-order modeling and Fourier priors into image restoration tasks. High-order models aim to capture more complex feature interactions within images, beyond the traditional pairwise relationships considered by many classical models. These higher-order dependencies are crucial in handling non-linear and intricate degradations (*e.g.*, low-light or motion blur) where simple pairwise relationships might fail to restore fine details. Recent research has focused on leveraging high-order interactions to enhance multi-modal image fusion. Zhou et al. (2025c) proposed a novel approach for multi-modal fusion by probing synergistic high-order interactions between different modalities. Similarly, Hou et al. (2024) introduced a linearly-evolved transformer, which refines the integration of high- and low-resolution multispectral images.

Fourier priors, which leverage the frequency domain of an image, have been effectively applied to restoration tasks, particularly in scenarios where degradation is global or exhibits frequency-specific patterns. Li et al. (2023) integrate Fourier domain priors into their image restoration models to suppress low-frequency noise and recover high-frequency details, showing significant improvements in tasks like deblurring and denoising. Similarly, Wang et al. (2022) used a Fourier-based approach to better model image degradation in low-light conditions, capturing and suppressing the undesired frequency components. These works collectively highlight the effectiveness of Fourier priors in addressing diverse restoration challenges, offering a promising avenue for further research in image enhancement.

3. Proposed SmoRestor

3.1. Architecture overview

Surgical image degradation often presents strong regional characteristics, such as localized smoke or blur, which pose significant challenges to uniform restoration strategies (refer to Fig. 2). Motivated by these observations, we design an efficient restoration framework for desmoking. As shown in Fig. 3, we first encode the smoky image and feed the features into a Fourier-prior-embedded frequency-decomposed regional transformer (equipped with an efficient KV-Cache mechanism). This design enables the integration of frequency-domain priors while maintaining low computational overhead. To further enhance learning, we leverage the Segment Anything (SAM) model through an auxiliary training branch guided by SAM-generated masks. This branch is discarded during inference, allowing the model to benefit from SAM's rich segmentation priors without incurring additional test-time costs. Finally, we incorporate an optical learning alignment module to alleviate spatial misalignments between the input and corresponding ground truth, further improving restoration quality.

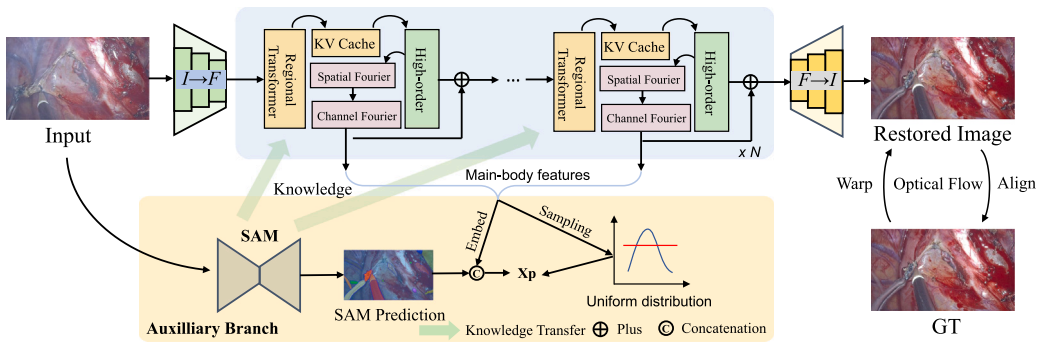

Fig. 3. The overall pipeline of the proposed surgical image desmoking framework.

3.2. Frequency-decomposed modeling

(1) **Regional Transformer.** Building on this concept, we introduce the regional transformer based on neighborhood attention. Given an input $x \in \mathbb{R}^{n\times d}$, where each row represents a d-dimensional token vector, and its linear projections Q, K, V, and relative positional biases $B(i,j)$, the attention weights for the ith token with a neighborhood size k are computed as the dot product of the ith token's query projection and the key projections of its k nearest neighbors:

$$A_k^i = \begin{bmatrix} Q_i K_{\rho_1(i)}^T + B(i,\rho_1(i)) \\ Q_i K_{\rho_2(i)}^T + B(i,\rho_2(i)) \\ \vdots \\ Q_i K_{\rho_k(i)}^T + B(i,\rho_k(i)) \end{bmatrix}, \tag{1}$$

where $\rho_j(i)$ denotes the jth nearest neighbor of i. The neighboring values V_k^i are defined as

$$V_k^i = \begin{bmatrix} V_{\rho_1(i)}^T \\ V_{\rho_2(i)}^T \\ \vdots \\ V_{\rho_k(i)}^T \end{bmatrix}^T. \tag{2}$$

Then, Neighborhood Attention for ith token is given by:

$$NA_k(i) = \mathrm{softmax}\left(\frac{A_k^i}{\sqrt{d}}\right) V_k^i, \tag{3}$$

where $\sqrt{d}$ is the scaling factor. This operation is repeated for every pixel in the feature map, where each pixel attends to a fixed window with padded inputs to handle edge cases. In our implementation, the k nearest neighbors are defined spatially within a fixed local window $\mathcal{N}_k(i) = \{j \mid |x_j - x_i| \le r, |y_j - y_i| \le r\}$, where $r = (k-1)/2$. Thus, the neighbor search is non-adaptive and performed implicitly through sliding-window indexing, not explicit pairwise distance computation. The total computational complexity is therefore reduced from the global $\mathcal{O}(n^2)$ to $\mathcal{O}(k^2 n)$, as each query token interacts only with its fixed-size neighborhood rather than all n tokens. Compared with Swin Transformer, which uses fixed-size (*e.g.*, 7×7) windows and suffers from boundary artifacts, our neighborhood attention-based regional transformer dynamically adjusts to local structure at the pixel level. This allows better preservation of fine-grained textures under surgical degradations. Moreover, while Swin exhibits quadratic complexity $\mathcal{O}(n^2)$, our method operates with $\mathcal{O}(k^2 \cdot n)$, making it more suitable for edge devices where hardware efficiency is crucial.

(2) **Frequency Decomposition.** We apply the Fourier Transform (FT) to analyze the frequency characteristics of surgical image degradation. By examining the phase and amplitude components in the Fourier domain, we find that the degradation effect primarily manifests in the amplitude. As shown in Fig. 2, swapping amplitude and phase between degraded and clean images shows that degradation can be decoupled from content in the Fourier domain. This supports our observation that amplitude serves as the main carrier of degradation. These findings motivate the use of Fourier-based priors for image restoration. We adopt the Fourier amplitude spectrum as a lightweight prior to separate degradation from anatomical content, providing informative preprocessing features for subsequent modeling. Empirically, exchanging amplitude and phase between degraded and clean images effectively transfers degradation patterns, which motivates this design. However, since real surgical smoke involves complex and dynamic interactions beyond amplitude, we integrate Fourier priors with higher-order neighborhood attention to enhance robustness.

Furthermore, we explicitly analyze the distinct roles of low- and high-frequency components. The low-frequency components mainly represent global illumination, color tone, and smoke density, which are strongly affected by haze-like attenuation. In contrast, the high-frequency components capture fine anatomical edges and texture details but are more sensitive to scattering noise and diffusion along tissue boundaries. These complementary characteristics motivate our Fourier Spatial Interaction and Fourier Channel Evolution modules to jointly balance global clarity and local sharpness during restoration.

The Frequency-decomposed Fourier block incorporates two key elements: Fourier Spatial Interaction and Fourier Channel Evolution. In contrast to prior works that rely on trigonometric functions (*e.g.*, sine and cosine), which are prone to numerical instability, we directly operate on the real and imaginary parts of the Fourier representation for greater robustness. To further enhance signal representation, we amplify the average channel response to emphasize the dominant spectral energy.

Spatial Fourier Modeling. Inspired by Zhou et al. (2023), Fourier spatial modeling first takes the feature maps as input and then applies the Fourier transform to convert the spatial features into real and imaginary components. As illustrated in Fig. 4, suppose that the features are denoted as $\mathbf{x} \in \mathbb{R}^{\mathrm{H}\times\mathrm{W}\times\mathrm{C}}$, the corresponding Fourier transform is expressed as

$$\mathbf{x}_{\mathcal{I}}^{(c)}, \mathbf{x}_{\mathcal{R}}^{(c)} = \mathcal{F}(\mathbf{x}^{(c)}), \tag{4}$$

where $\mathrm{c} = 1, \ldots, \mathrm{C}$, $\mathbf{x}_{\mathcal{I}}^{(c)}$ and $\mathbf{x}_{\mathcal{R}}^{(c)}$ indicate the real and imaginary respectively. Then, we implement the spatial interaction by a stack of depth-wise convolution with the kernel size of 3×3 and the ReLU function. Each channel is processed independently, and the same depth-wise kernel is applied to both real and imaginary parts. The spatial interaction can be written as follows:

$$\mathbf{s}_{\mathcal{I}}^{(c)} = \mathtt{ReLU} \circ \mathtt{DW}^{(c)}(\mathbf{x}_{\mathcal{I}}^{(c)}), \tag{5}$$

$$\mathbf{s}_{\mathcal{R}}^{(c)} = \mathtt{ReLU} \circ \mathtt{DW}^{(c)}(\mathbf{x}_{\mathcal{R}}^{(c)}), \tag{6}$$

where ReLU and DW indicate the ReLU function and depth-wise convolution respectively. After the convolutional interaction, the enhanced

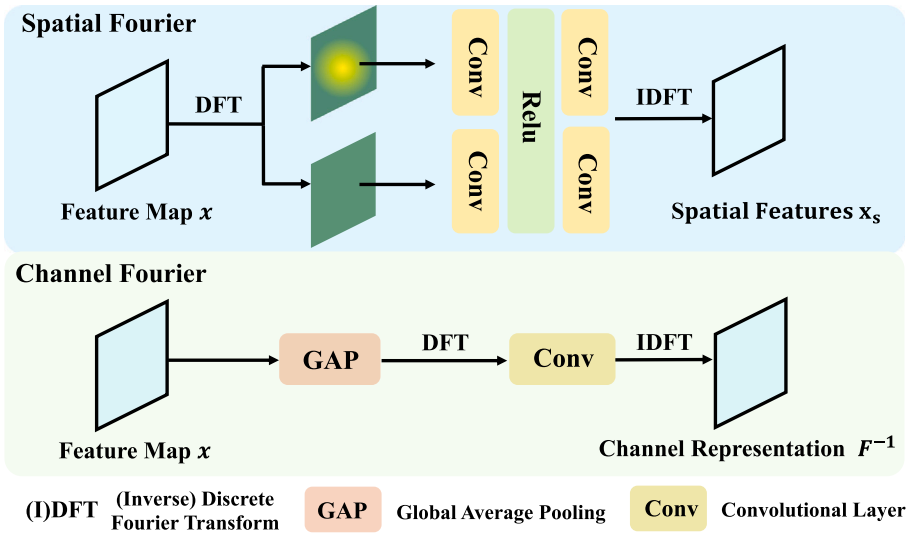

Fig. 4. The proposed Spatial Fourier and Channel Fourier module in SmoRestor.

real and imaginary components are transformed back into the spatial domain via inverse Fourier Transform:

$$\mathbf{x}_s^{(c)} = \mathcal{F}^{-1}(\mathbf{s}_{\mathcal{I}}^{(c)}, \mathbf{s}_{\mathcal{R}}^{(c)}). \tag{7}$$

Finally, we combine the Fourier-transformed spatial features $\mathbf{x}_s$ by concatenating each component $\mathbf{x}$ with the spatial features processed by a half-instance normalization block, resulting in the final output.

Channel-wise Fourier Modeling. As illustrated in Fig. 4, given a feature map $\mathbf{x} \in \mathbb{R}^{\mathrm{H}\times\mathrm{W}\times\mathrm{C}}$, we first apply global average pooling (GAP) to summarize each channel's spatial information:

$$\mathtt{GAP}(\mathbf{x}) = \frac{1}{\mathrm{H}\times\mathrm{W}} \sum_{h=0}^{\mathrm{H}-1} \sum_{w=0}^{W-1} F(h, w, c). \tag{8}$$

This operation computes the average value of each channel over the entire spatial domain, resulting in a C-dimensional vector. To leverage frequency-domain characteristics, we perform a *channel-wise Fourier Transform* on the pooled vector. This transform decomposes each channel into real and imaginary frequency components, effectively encoding both high-frequency details (*e.g.*, edges) and low-frequency structures (*e.g.*, smooth regions), which are crucial for accurate image restoration. For channel C in the feature map $F \in \mathbb{R}^{\mathrm{H}\times\mathrm{W}\times\mathrm{C}}$, the Fourier Transform $\mathcal{F}(\mathtt{GAP}(\mathbf{x}))$ is

$$\mathcal{F}(\mathrm{z}) = \sum_{c=0}^{\mathrm{C}-1} \mathtt{GAP}(\mathbf{x}|c) \cdot e^{-2\pi jcz/C}, \tag{9}$$

$$\mathbf{C}_{\mathcal{I}} = \mathtt{Relu} \circ \mathtt{DW}_1(\mathcal{F}_{\mathcal{I}}(\mathrm{z})), \tag{10}$$

$$\mathbf{C}_{\mathcal{R}} = \mathtt{Relu} \circ \mathtt{DW}_1(\mathcal{F}_{\mathcal{R}}(\mathrm{z})). \tag{11}$$

The Inverse Fourier Transform (IFT) is then applied to convert the filtered frequency components back to the original channel representation:

$$\mathcal{F}^{-1}(\mathrm{C}) = \frac{1}{\mathrm{C}} \sum_{\mathrm{c}=0}^{\mathrm{C}-1} (\mathbf{C}_{\mathcal{R}} + \mathrm{j}\mathbf{C}_{\mathcal{I}}) \cdot \mathrm{e}^{2\pi \mathrm{jcz}/\mathrm{C}}. \tag{12}$$

This process preserves essential frequency-domain cues. When combined with GAP, it enhances the model's ability to capture both fine details and global structure.

(3) **High-order Design.** We build on the interpretation of QK attention as a first-order operator under the softmax framework, where it acts as a distributional weighting over the value matrix. This view allows self-attention to be reformulated as a first-order expectation over the value features, offering a more principled understanding of its statistical behavior. Based on this insight, we replace the costly cascade of softmax-based attention blocks (which incur $\mathcal{O}(kn^2)$ complexity) with a high-order functional formulation using 1D convolutions. This design achieves comparable expressiveness while reducing computational cost to linear complexity. In the context of spatial correlation calculation, the attention mechanism NA_k is expressed as

$$\mathbf{a}_{i,j} = \sum_{m=0}^{C} \sum_{n=0}^{C} \frac{\mathbf{q}_{i,m}\mathbf{k}_{n,j}}{\sum_{i=1}^{C} \mathbf{q}_{i,m}\mathbf{k}_{n,j}}, \tag{13}$$

where $\mathbf{q}_{i,m} \in \mathbb{Q}$, $\mathbf{k}_{n,j} \in \mathbb{K}$, and $\mathbf{a}_{i,j} \in \mathbb{A}$. Given the output feature map $\mathbf{O} \in \mathbb{R}^{H\times W\times C}$, the attention output can be written as:

$$\mathbf{O}_{k,s} = \sum_{m=0}^{C} \sum_{n=0}^{C} \mathbf{a}_{l,m}\mathbf{v}_{n,s} = \mathcal{O}_1(\mathbf{V}), \tag{14}$$

where $\mathcal{O}_1(\cdot)$ denotes the first-order weights. To better understand the statistical behavior of attention, we reshape spatial features into column vectors $\mathbf{k}^i$ and $\mathbf{q}^j$, and express the attention output as:

$$\begin{aligned} &\mathbf{O}_j = \mathrm{softmax}(\mathbf{q}_i, \mathbf{k}_j) \cdot \mathbf{v}_j, \quad \forall i, \\ &0 < \mathrm{softmax}(\mathbf{q}_i, \mathbf{k}_j) < 1, \\ &\sum_i \mathrm{softmax}(\mathbf{q}_i, \mathbf{k}_j) = 1, \quad \forall i. \end{aligned} \tag{15}$$

Specifically,

$$\mathrm{softmax}(\mathbf{k}_i, \mathbf{q}_j) = \frac{\exp((\mathbf{q}_i)^T)\mathbf{k}_j}{\sum_{l=1}^{C} \exp((\mathbf{q}_i)^T)\mathbf{k}^l}. \tag{16}$$

This calculation serves as a normalization function that transforms $\exp((\mathbf{q}^i)^T\mathbf{k}^1), \ldots, \exp((\mathbf{q}^i)^T\mathbf{k}^C)$ into probabilities. Mathematically, for any function $p(x)$ satisfying the conditions $0 \le p(x) \le 1$ and $\sum_x p(x) = 1$, its first-order statistic calculation is equivalent to:

$$p(\mathbf{v}_j) \propto \mathrm{softmax}(\mathbf{q}_i, \mathbf{k}_j), \tag{17}$$

$$\mathbf{O}_j = \int_0^1 p(\mathbf{v}_j)\mathbf{v}_j d\mathbf{v} \approx \mathbb{E}(\mathbf{v}_j). \tag{18}$$

Here, $\mathbb{E}(\cdot)$ represents the first-order expectation. This insight allows us to replace traditional second-order interaction sequences with more efficient high-order modeling through attention sharing.

Considering the first-order statistical expectation of a variance tensor, we assume a probability distribution $p(\mathbf{v})$ that satisfies the following constraints:

$$0 \le p(\mathbf{v}) \le 1, \quad \sum_i p(\mathbf{v}) = 1. \tag{19}$$

High-Order : 고차(차수) ...
→ first-Order : 일차 ...

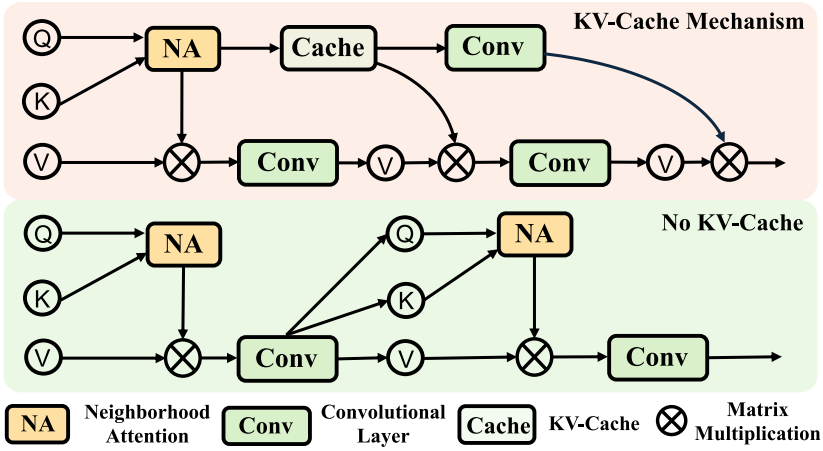

Fig. 5. The detailed structure of our KV-cache mechanism in SmoRestor.

Thus, the expectation of $\mathbf{v}_j$ can be expressed as:

$$\mathbb{E}(\mathbf{v}_j) = \int_0^1 p(\mathbf{v})\mathbf{v}_j d\mathbf{v}. \tag{20}$$

By considering softmax(·) as a special case of a probability distribution, we deduce that self-attention fundamentally operates as a first-order linear weight function and can be represented by a 1D convolution.

Building on this formulation, we propose a high-order attention mechanism that encodes spatial dependencies using recursive convolutions. To address saturation in repeated attention blocks, we adopt a recursive update strategy, where each attention map $\mathbf{A}^i$ is derived from its predecessor via a 1×1 convolution:

$$\mathbf{A}^i = \mathrm{Conv}_{1\times1}(\mathbf{A}^{i-1}), \tag{21}$$

where $\mathbf{A}^{i-1}$ and $\mathbf{A}^i$ refer to adjacent attention calculations. From the above analysis, each QK-attention block can be formally regarded as a first-order operator $F_1(X) = \mathrm{Softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$, where $X \in \mathbb{R}^{n\times d}$. When stacked, these operators form a composite mapping $F^{(k)}(X) = F_k \circ F_{k-1} \circ \cdots \circ F_1(X)$, which naturally introduces higher-order dependencies among tokens. This explains the expressive power of cascaded attention, but also its quadratic complexity. To alleviate this, we structurally substitute repeated QK computations with a 1D convolution $C_i(X) = \mathrm{Conv1D}(X;\theta_i)$, which approximates local correlations at $\mathcal{O}(n)$ cost. Through multi-layer composition, the convolutional form retains high-order representation ability while being significantly more efficient. This theoretical view underpins our higher-order Fourier-embedded neighborhood attention, which integrates Fourier priors with efficient high-order modeling.

(4) **Efficient QK-sharing Mechanism.** To accelerate inference while preserving modeling capacity, we propose a KV-cache-inspired mechanism specifically designed for surgical image restoration. Unlike conventional KV-caching in autoregressive models, which focuses on temporal reuse, our approach emphasizes the reuse of QK attention maps across spatial tokens.

This design is motivated by the spatial continuity and low-rank structure often observed in surgical features. As illustrated in Fig. 5, rather than recomputing QK interactions at each decoding step or across transformer layers, we precompute and share QK attention weights throughout the network. To further reduce computational overhead, we approximate the QK interactions using localized 1D convolutions, which capture essential spatial correlations while maintaining efficiency, enabling efficient attention computation without sacrificing restoration quality, achieving a practical balance between speed and accuracy.

(5) **Complexity Analysis.** Our pipeline is a single-pass deterministic network. The dominant costs are: (i) 2D FFT/IFFT at $\mathcal{O}(HW\log(HW))$, (ii) neighborhood attention with window k at $\mathcal{O}(HWk^2)$, (iii) the high-order 1D-convolutional formulation at $\mathcal{O}(HW)$, and (iv) light QK sharing via KV-cache that avoids repeated QK products. In contrast, diffusion-based restoration requires S reverse steps, so runtime scales as $T_{\text{diff}} \approx S \cdot T_{\text{unet-step}}$, with memory proportional to the per-step UNet and feature activations. Therefore, our total runtime $T_{\text{ours}}(H,W)$ grows sub-quadratically with resolution and *does not* incur step-wise multiplicative cost, making it favorable for real-time surgical settings.

3.3. Large model training-testing distilling mechanism

We leverage the Segment Anything Model (SAM) (Kirillov et al., 2023) to extract semantic priors from degraded images. These priors help restoration models better distinguish meaningful anatomical structures from ambiguous degradations, thereby enhancing restoration performance. Given a degraded image $\mathbf{I}$, we can utilize the automatic mode of SAM to obtain the segmentation mask:

$$\mathbf{S}_{AM} = f_{\texttt{SAM}}(\mathbf{I}), \tag{22}$$

where $\mathbf{S}_{AM}$ denotes the set of semantic masks extracted by SAM, and $f_{\texttt{SAM}}$ is the SAM segmentation function. To balance the powerful guidance of large models during training with the need for efficient inference, we propose an innovative learning paradigm that transfers knowledge from SAM to the main-body structure.

(1) Main Branch Feature Extraction. The input image $\mathbf{I}$ is passed through the main network $f_{\texttt{main}}$ to obtain the set of feature maps $\mathbf{x}_1 \ldots \mathbf{x}_l$:

$$\mathbf{x}_1, \ldots, \mathbf{x}_l = f_{\texttt{main}}(\mathbf{I}). \tag{23}$$

(2) Random Mask Generation. A random binary mask $\mathbf{M} \sim P(\mathbf{M})$ is sampled from a predefined distribution P, such as a Bernoulli distribution. The masked feature is computed as:

$$\mathbf{x}^p = \texttt{Sampling}(\mathbf{x}_1, \ldots, \mathbf{x}_l), \tag{24}$$

$$\mathbf{x}^p_{\texttt{mask}} = \mathbf{x}^p \odot \mathbf{M}, \tag{25}$$

where $\odot$ and `Sampling` represent element-wise multiplication and random sampling. To ensure a diverse semantic supervision across hierarchical representations, a feature map $\mathbf{x}^p$ is randomly selected from one of three stages. Before masking, the feature is aligned in channel dimension via a 1×1 projection. A random Bernoulli mask $\mathbf{M} \sim \mathrm{Bernoulli}(p)$ with keep probability $p = 0.5$ is applied to simulate partial feature corruption. Simultaneously, the SAM segmentation image is encoded by a shallow 3×3 convolution to generate semantic guidance, which provides the anatomical prior used in reconstruction.

(3) Auxiliary Branch Feature Reconstruction. The masked feature $\mathbf{x}^p_{\mathtt{mask}}$ concatenated by semantic mask $\mathbf{S}_{AM}$ is passed into the auxiliary branch $f_{\mathtt{aux}}$ to reconstruct the original feature $\hat{\mathbf{x}}^p$:

$$\hat{\mathbf{x}}^p = f_{\mathtt{aux}}(\mathbf{x}^p_{\mathtt{mask}}, \mathbf{S}_{AM}). \tag{26}$$

(4) Training-test Strategy. Finally, the reconstruction loss $\mathcal{L}_{\mathtt{dis}}$ is computed as the mean squared error (MSE) between the reconstructed and original features:

$$\mathcal{L}_{\mathtt{dis}} = \mathbb{E}\left[\|\hat{\mathbf{x}}^p - \mathbf{x}^p\|_2^2\right]. \tag{27}$$

The random masking acts as a regularized teacher–student constraint: when partial features are occluded, the model must rely on $\mathbf{S}_{\mathrm{AM}}$ to recover the missing context. Hence, SAM masks serve not as deterministic spatial weights but as semantic complements guiding feature completion. This coupling of random masking and SAM guidance encourages robust feature distillation, allowing the network to learn both spatial attention and structural priors in a unified manner.

3.4. Loss function

Regarding the misalignment between the restoration output and supervision, inspired by Wu et al. (2024), a pre-trained optical flow network O_f is first used to estimate the optical flow $\Psi_{\mathtt{G}\to y}$ from $\mathtt{G}$ to y:

$$\Psi_{\mathtt{G}\to y} = O_f(\mathtt{G}, y). \tag{28}$$

Then, $\mathtt{G}$ is back warped towards y with a warping operation W according to the estimated optical flow $\Psi_{\mathtt{G}\to y}$:

$$\hat{G}_{G\to y} = W(G, \Psi_{\mathtt{G}\to y}). \tag{29}$$

Finally, $\hat{G}_{G\to y}$ is spatially aligned with y. The reconstruction loss $L_{\mathtt{rec}}$ of the restoration model can be written as

$$\mathcal{L}_{\mathtt{rec}} = \sum_{i=1}^{N} \left| V_i \odot (\hat{G}_{G\to y} - y) \right|_1, \tag{30}$$

where $\odot$ is a pixel-wise multiplication operation. V_i is a mask that indicates the valid positions of optical flow. The jth element of V_i can be calculated as

$$V_{ij} = \mathrm{sgn}\left(\max\left(0, \left[W(1, \Psi_{\mathtt{G}\to y})\right]_j - \tau\right)\right), \tag{31}$$

where W is the warping operation, max and sgn are the maximum and sign functions, respectively. 1 is an all-1 matrix, $[\cdot]_j$ denotes the jth element, and τ is set to 0.999.

The final training objective combines the spatially aligned reconstruction loss and the auxiliary distillation loss. The total loss is defined as:

$$\mathcal{L}_{\mathtt{total}} = \lambda_{\mathtt{rec}}\mathcal{L}_{\mathtt{rec}} + \lambda_{\mathtt{dis}}\mathcal{L}_{\mathtt{dis}}, \tag{32}$$

where $\lambda_{\mathtt{rec}}$ and $\lambda_{\mathtt{dis}}$ are weighting coefficients for the reconstruction and distillation losses, respectively. By jointly optimizing these objectives, our model benefits from both fine-grained pixel alignment and knowledge transfer from large models.

4. Experiments

4.1. Datasets

Several datasets (Venkatesh et al., 2020; Pan et al., 2022; Xia et al., 2024b; Pei et al., 2025a) derived from real endoscopic surgeries have been proposed for addressing the task of surgical image desmoking. Among the available resources, Desmoke-LAP (Pan et al., 2022) focuses on hysterectomy videos, while DesmokeData (Xia et al., 2024b) originates from laparoscopic radical prostatectomy procedures acquired using the da Vinci Si robotic surgical system. Since Desmoke-LAP lacks paired clean labels required for model training, we select DesmokeData to train and test both the proposed SmoRestor and comparative models under fully supervised conditions. Specifically, DesmokeData consists of 21 video sequences and 961 paired samples of degraded images and corresponding clean labels, each with a resolution of 700 × 350 pixels. All samples are divided into 761 pairs for training and 200 pairs for testing through random sampling. To examine the model's performance in real-world clinical surgery, we also collect 200 pairs of private in-vivo data from real thoracoscopic right lung segment resection surgeries for direct testing without any additional fine-tuning. The paired GT labels are obtained following the manner described in Xia et al. (2024b) for performance evaluation.

4.2. Evaluation metrics

Similar to the general evaluation metrics for image restoration tasks, the model performance is evaluated by Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS) metrics. PSNR measures the pixel-level fidelity between the restored image I and the ground truth $\hat{I}$, which is computed as:

$$\mathrm{PSNR}(I, \hat{I}) = 10 \cdot \log_{10}\left(\frac{\mathrm{MAX}^2}{\mathrm{MSE}}\right), \tag{33}$$

where MAX denotes the maximum possible pixel value (255 for 8-bit images). A higher PSNR value indicates better reconstruction quality with lower pixel-level distortion. MSE is the mean squared error defined by:

$$\mathrm{MSE} = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}(I_{ij} - \hat{I}_{ij})^2, \tag{34}$$

with H and W representing image height and width, respectively. SSIM evaluates structural similarity between the restored image and the reference by considering luminance, contrast, and structural components. It is calculated as:

$$\mathrm{SSIM}(I, \hat{I}) = \frac{(2\mu_I\mu_{\hat{I}} + C_1)(2\sigma_{I\hat{I}} + C_2)}{(\mu_I^2 + \mu_{\hat{I}}^2 + C_1)(\sigma_I^2 + \sigma_{\hat{I}}^2 + C_2)}, \tag{35}$$

where μ and σ represent mean and variance of image intensities, $\sigma_{I\hat{I}}$ is the covariance, and C_1, C_2 are small constants for numerical stability. SSIM ranges from 0 to 1, with values closer to 1 indicating higher perceptual similarity. LPIPS is a perceptual metric that leverages deep network features to measure the similarity of local image patches in a learned feature space rather than pixel space. A lower LPIPS score signifies better perceptual quality.

Referring to Wu et al. (2024), we also adopt non-reference metrics, *i.e.*, Naturalness Image Quality Evaluator (NIQE) and Perceptual Index (PI), to evaluate the restored results. NIQE fits the statistical features of natural scenes using a multivariate Gaussian model in the spatial domain and calculates the distance between the restored image statistics and the naturalness model. Lower NIQE values indicate more natural-looking images. PI combines NIQE with the Ma score (Gao et al., 2015) to produce a unified score as:

$$\mathrm{PI} = 0.5 \times \big((10 - \mathrm{Ma}) + \mathrm{NIQE}\big), \tag{36}$$

where lower PI values indicate higher perceptual quality.

4.3. Implementation details

Our model is implemented with PyTorch (Paszke et al., 2017) and trained on two NVIDIA RTX 4090 GPUs with the Adam optimizer. During training, the input image is randomly cropped into 128 × 128 patches with a batch size of 2 and a total of 200,000 iterations. The initial learning rate of our model is set to 1e−4, and the learning rate is decreased by half every 100k iterations throughout the training process. In addition, we use random crop and horizontal flip as data augmentation. Notably, all comparison models use a pre-trained PWC-Net (Sun et al., 2018) for pairing unaligned GT with restored images.

Table 1
Quantitative comparison for desmoking with representative models on the DesmokeData dataset.

Methods	PSNR ↑	SSIM ↑	LPIPS ↓	NIQE ↓	PI ↓	Params (M) ↓	GFLOPs ↓	FPS ↑
ConvIR (Cui et al., 2024)	24.0991	0.8526	0.1579	4.0123	8.9436	14.83	3223	2.28
FocalNet (Cui et al., 2023)	24.5770	0.8810	0.1576	3.8045	38.0652	3.74	362	**24.28**
Fourmer (Zhou et al., 2023)	23.0514	0.7943	0.1975	4.3301	29.1286	0.68	502	5.51
Restormer (Zamir et al., 2022)	23.3150	0.8461	0.1585	3.9654	7.3421	26.13	1673	1.29
MambaIR (Guo et al., 2024a)	22.4050	0.7681	0.2314	4.5283	19.6712	1.31	2021	0.37
Histoformer (Sun et al., 2025)	24.0344	0.6539	0.3125	5.2344	10.5523	29.92	1994	0.59
RAMiT (Choi et al., 2024)	23.5030	0.8227	0.1561	4.1055	38.3316	**0.30**	**164**	3.48
AMIR (Yang et al., 2024a)	22.4637	0.8649	0.1855	4.2270	8.8941	23.54	1509	1.16
Restore-RWKV (Yang et al., 2024c)	21.8635	0.8017	0.1844	4.3388	9.0782	28.44	1780	0.48
AST (Zhou et al., 2024)	23.8542	0.8285	0.1742	4.1125	9.4357	19.92	1280	0.95
X-Restormer (Chen et al., 2024b)	23.5216	0.8193	0.1864	4.3012	10.8235	42.52	1985	0.72
SFHformer (Jiang et al., 2024)	23.7654	0.8321	0.1728	4.0251	7.9814	7.67	540	6.12
SmoRestor(Ours)	**24.6192**	**0.8834**	**0.1496**	**3.7520**	**6.9825**	15.32	1153	3.46

4.4. Comparisons

We validate the effectiveness and generalizability of SmoRestor through extensive experiments on DesmokeData (Xia et al., 2024b) and our private in-vivo dataset.

4.4.1. Comparisons on DesmokeData

To evaluate the generalization ability of the proposed SmoRestor, we conduct comprehensive comparisons on the DesmokeData dataset (Xia et al., 2024b), which contains 961 noisy laparoscopic images with corresponding aligned ground truth. We compare the proposed method with nine representative generic and medical image restoration models. These include CNN-based (ConvIR (Cui et al., 2024), FocalNet (Cui et al., 2023)), transformer-based (Restormer (Zamir et al., 2022), X-Restormer (Chen et al., 2024b), AST (Zhou et al., 2024), RAMiT (Choi et al., 2024), Histoformer (Sun et al., 2025)), state-space and Fourier-based methods (MambaIR (Guo et al., 2024a), Fourmer (Zhou et al., 2023), SFHformer (Jiang et al., 2024)), and medical-specific models (AMIR (Yang et al., 2024a), Restore-RWKV (Yang et al., 2024c)).

Quantitative Comparisons. Table 1 summarizes the quantitative performance of our method and nine representative baselines on DesmokeData in terms of PSNR, SSIM, LPIPS, NIQE, and PI. As illustrated in Table 1, our method achieves the best overall performance, attaining the highest PSNR (24.6192), the highest SSIM (0.8834), the lowest LPIPS (0.1496), the lowest NIQE (3.7520), and the lowest PI (6.9825), thereby confirming its strong capability in restoring clean, structurally faithful, and perceptually realistic images.

Compared to strong CNN-based models such as ConvIR (24.0991 PSNR) and FocalNet (24.5770 PSNR), our method demonstrates clear advantages, particularly in SSIM and LPIPS, indicating better preservation of fine structures and perceptual quality. Notably, our model surpasses the widely used Restormer (Zamir et al., 2022), while also achieving a significantly lower LPIPS. Transformer-based and state-space models, including RAMiT, MambaIR, and Restore-RWKV, underperform in comparison, likely due to their limited capability in capturing low-level texture patterns that are crucial in medical image enhancement. For instance, MambaIR yields only 22.4050 in PSNR and exhibits the highest LPIPS score (0.2314), revealing its inferior perceptual restoration. Similarly, Histoformer, despite being designed for medical restoration, struggles in desmoking scenarios, resulting in the lowest SSIM (0.6539) among all competitors.

In contrast, the superior performance of our model stems from its regional attention mechanism and the incorporation of Fourier-prior features, which jointly enhance both the fidelity and perceptual consistency of restored images. These results confirm that SmoRestor not only achieves superior performance under standard distortion and perceptual metrics but also sets a new benchmark for laparoscopic desmoking tasks.

Qualitative Comparisons. We also provide more qualitative results of our model and other well-known models on DesmokeData (Xia et al., 2024b) dataset. As shown in Fig. 6, our method recovers surgical noise images with better quality compared to other competitors, which further demonstrates the robustness of our method for desmoking.

4.4.2. Evaluation on our in-house dataset

To further validate the effectiveness and practical applicability of our method, we also conduct a comparison experiment on our private in-vivo dataset from real thoracoscopic surgeries.

Quantitative Comparisons. Table 2 presents the quantitative comparison results for desmoking on our dataset. Compared to existing methods, the proposed framework achieves superior performance, demonstrating its strong generalization capability for surgical desmoking. Specifically, our approach improves by about 0.2 over the second-ranked Restormer (Zamir et al., 2022) on PSNR, and consistently surpasses other models on SSIM, NIQE, and PI metrics. In comparison with other medical image restoration models (*i.e.*, AMIR (Yang et al., 2024a), Restore-RWKV (Yang et al., 2024c)), the proposed SmoRestor obtains a lower PI score in non-reference metrics, indicating that our method excels at restoring images with higher naturalness. In addition, the proposed paradigm maintains moderate model parameters and computational costs while attaining high-quality predictions using large vision models.

Visual Comparisons. We also conduct a qualitative comparison of representative methods for desmoking in Fig. 7. We can see that our solution achieves promising performance in restoring low-quality regions and fine details, especially for key operating regions. In challenging endoscopic scenarios, generic image restoration models tend to have minimal effect in restoring noisy locations. In contrast, our approach effectively restores both the image structure and details, mitigating the interference from noisy regions and enhancing visual clarity for surgeons during the operation process.

4.4.3. Cross-validation on DesmokeData and in-house dataset

As shown in Table 3, the cross-validation results between DesmokeData (Xia et al., 2024a) and in-house dataset demonstrate the strong generalizability of our proposed SmoRestor. When trained on DesmokeData and tested on in-house dataset (Exp 1), SmoRestor achieves the best overall performance, surpassing all competing methods by noticeable margins in PSNR and SSIM while maintaining lower perceptual metrics (LPIPS, NIQE, and PI). Similarly, under the inverse setting (Exp 2), where models are trained on in-house dataset and tested on DesmokeData, our method consistently outperforms others, achieving a PSNR of 24.21 dB and SSIM of 0.870, reflecting robust adaptation to unseen surgical domains. These results suggest that SmoRestor effectively captures domain-invariant degradation patterns through its high-order Fourier modeling and SAM-guided distillation, enabling reliable desmoking even across datasets with distinct visual and surgical characteristics. Overall, the superior cross-dataset results highlight the strong generalization and transferability of SmoRestor in real-world surgical restoration scenarios.

4.5. Ablation studies

To verify the impact of each component on the performance of the proposed SmoRestor, we conduct a comprehensive ablation experiment

Table 2
Quantitative comparison for surgical image desmoking with representative models on in-house dataset.

Methods	PSNR ↑	SSIM ↑	LPIPS ↓	NIQE ↓	PI ↓	Params (M) ↓	GFLOPs ↓	FPS ↑
ConvIR (Cui et al., 2024)	18.4640	0.6492	0.3993	5.8070	42.0036	14.83	3223	2.28
FocalNet (Cui et al., 2023)	18.3470	0.6545	0.3970	5.6307	36.6765	3.74	362	**24.28**
Fourmer (Zhou et al., 2023)	18.2081	0.6421	0.4084	5.4177	28.1543	0.68	502	5.51
Restormer (Zamir et al., 2022)	18.4700	0.6335	**0.3678**	5.3732	11.8538	26.13	1673	1.29
MambaIR (Guo et al., 2024a)	18.4557	0.6550	0.4305	6.8445	49.5051	1.31	2021	0.37
Histoformer (Sun et al., 2025)	16.6871	0.3764	0.6402	6.9394	13.2016	29.92	1994	0.59
RAMiT (Choi et al., 2024)	18.3475	0.6466	0.3858	5.8406	29.8502	**0.30**	**164**	3.48
AMIR (Yang et al., 2024a)	18.0971	0.6365	0.4091	6.3321	17.1409	23.54	1509	1.16
Restore-RWKV (Yang et al., 2024c)	17.8753	0.6485	0.3977	5.8851	36.2842	28.44	1780	0.48
AST (Zhou et al., 2024)	18.4614	0.6163	0.4254	5.4373	21.3512	19.92	1280	0.95
X-Restormer (Chen et al., 2024b)	18.0312	0.6224	0.4585	5.5388	32.2105	42.52	1985	0.72
SFHformer (Jiang et al., 2024)	17.9382	0.6254	0.4092	5.6831	10.8132	7.67	540	6.12
SmoRestor(Ours)	**18.6708**	**0.6587**	0.3864	**5.3033**	**10.6657**	15.32	1153	3.46

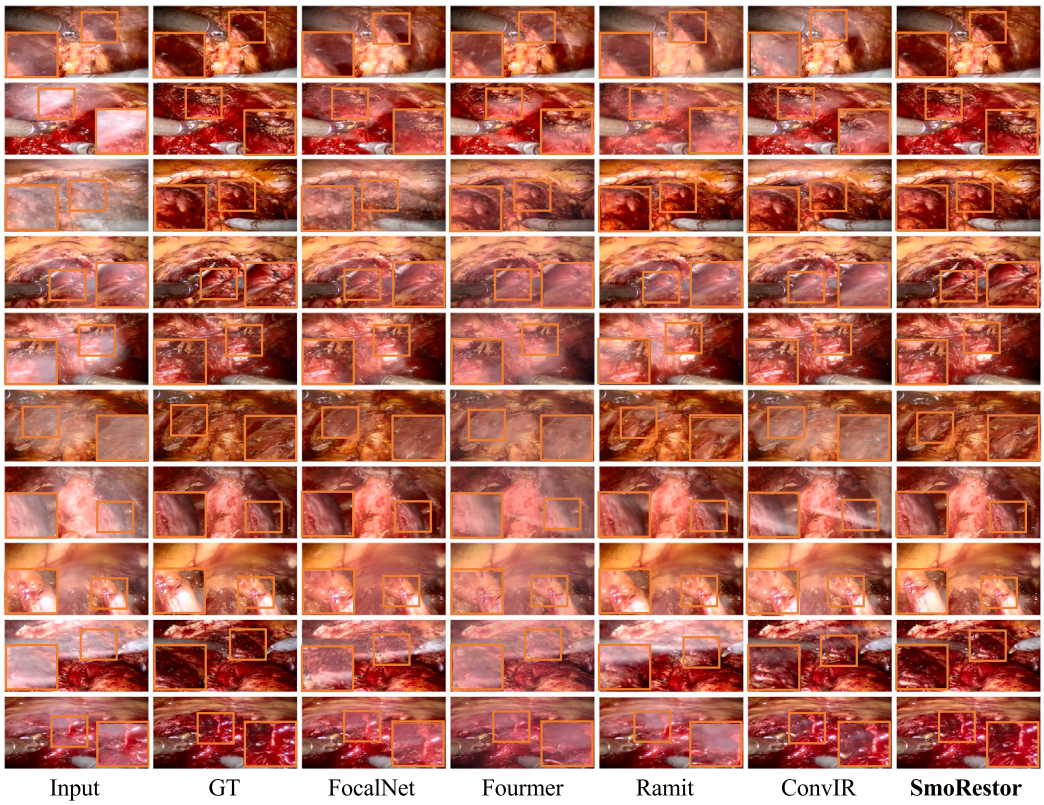

Fig. 6. Additional visualizations of the proposed method and other representative models on DesmokeData (Xia et al., 2024b) dataset. Zoom in for details.

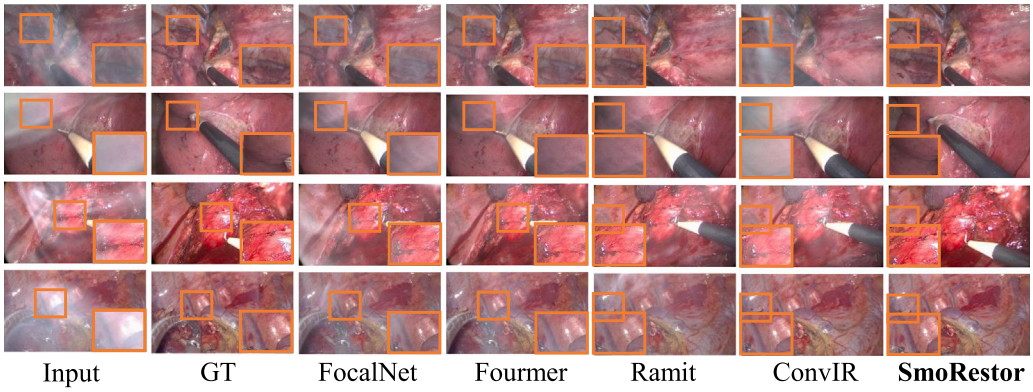

Fig. 7. Additional visualizations of the proposed method and other representative models on our in-house dataset. Zoom in for details.

Table 3
Cross validation for desmoking between DesmokeData (Xia et al., 2024a) and in-house dataset.

Tasks	Methods	Training set	Test set	PSNR ↑	SSIM ↑	LPIPS ↓	NIQE ↓	PI ↓
Exp 1	ConvIR (Cui et al., 2024)	DesmokeData	In-house Data	15.55	0.514	0.446	5.317	19.29
	FocalNet (Cui et al., 2023)			16.33	0.598	0.368	4.82	14.24
	Restormer (Zamir et al., 2022)			14.36	0.518	0.416	5.263	18.23
	MambaIR (Guo et al., 2024a)			15.43	0.563	0.391	5.735	33.19
	Histoformer (Sun et al., 2025)			15.56	0.556	0.423	5.221	7.94
	RAMiT (Choi et al., 2024)			14.93	0.543	0.384	5.129	18.82
	AMIR (Yang et al., 2024a)			14.91	0.541	0.406	5.278	15.36
	SmoRestor (Ours)			**16.52**	**0.622**	**0.356**	**4.73**	**13.97**
Exp 2	ConvIR (Cui et al., 2024)	In-house Data	DesmokeData	23.54	0.836	0.163	4.198	10.03
	FocalNet (Cui et al., 2023)			23.91	0.858	0.160	3.923	9.518
	Restormer (Zamir et al., 2022)			22.76	0.822	0.161	4.466	10.92
	MambaIR (Guo et al., 2024a)			21.93	0.751	0.234	5.38	14.62
	Histoformer (Sun et al., 2025)			23.37	0.639	0.317	6.179	16.47
	RAMiT (Choi et al., 2024)			23.01	0.805	0.152	4.307	10.58
	AMIR (Yang et al., 2024a)			21.92	0.846	0.189	5.093	12.64
	SmoRestor (Ours)			**24.21**	**0.870**	**0.147**	**3.88**	**9.34**

Table 4
Ablations for main components in our framework on the DesmokeData (Xia et al., 2024b) dataset for desmoking, including High-order modeling (High-order), Fourier prior (Fourier), and Large vision model distilling mechanism (LVM).

Models	High-order	Fourier	LVM	PSNR ↑	SSIM ↑	LPIPS ↓	FPS ↑
M.1				17.5220	0.5983	0.4112	**5.78**
M.2		✓	✓	20.1653	0.7316	0.2398	3.98
M.3	✓		✓	21.4126	0.7722	0.2135	4.23
M.4	✓	✓		22.0371	0.8024	0.1827	4.62
M.5	✓	✓	✓	**24.6192**	**0.8834**	**0.1496**	3.75

on the DesmokeData (Xia et al., 2024b) dataset. We conducted the main experiments on three core components for the desmoking task, including high-order modeling, Fourier prior, and knowledge transfer from large vision models. In addition, we further explore the effects of segmentation priors, transformer block depth, alignment-aware reconstruction loss, and frequency-decomposed attention, all of which contribute to enhancing the restoration performance under various challenging conditions. These ablations provide a more comprehensive understanding of how each component influences the effectiveness, efficiency, and generalization of the proposed SmoRestor.

4.5.1. Ablations for Fourier prior

The Fourier prior is designed to decompose both the context and degradation in the Fourier space, where it becomes significantly easier to remove smoke. By leveraging this prior, we are able to enhance the model's ability to recover high-quality information. To evaluate its effects, we investigate the influence of removing the Fourier prior on model performance. Comparing the results in the third and fifth rows of Table 4, it becomes clear that the inclusion of this module substantially improves the model performance, as evidenced by the increase in the PSNR metric. This comparison not only reinforces the efficacy of the Fourier prior but also underscores its positive contribution to improving the overall performance of our framework.

4.5.2. Effect of high-order modeling

To thoroughly evaluate the effectiveness of our high-order modeling, we replace the high-order design with the widely adopted ResNet block in the proposed framework. This comparison allows us to isolate the impact of the high-order design on model performance. As shown in the second row of Table 4, when high-order modeling is removed, our model exhibits a noticeable degradation in performance. This result demonstrates that the inclusion of high-order modeling significantly enhances the model's ability to capture and process complex, interdependent information. The comparison further highlights the crucial role that the high-order design plays in improving the global modeling capability of our model, underscoring its importance in achieving superior performance.

Table 5
Effect of different knowledge priors in our auxiliary branch for desmoking.

Models	PSNR ↑	SSIM ↑	LPIPS ↓
No Semantic mask	22.0371	0.8024	0.1827
MobileSAM (Zhang et al., 2023)	23.0921	0.8434	0.1715
MedSAM (Ma et al., 2024)	24.2189	0.8651	0.1613
SAM (Kirillov et al., 2023) (default)	**24.6192**	**0.8834**	**0.1496**

4.5.3. Impact of large vision model distillation

Embedding pre-trained large vision models directly into our model significantly impacts the computational cost of the network, especially when performing inference on edge devices with limited memory. To this end, we propose an efficient knowledge transfer strategy for large vision model distillation. To investigate the impact of the distillation scheme on the performance of our model, we also ablate this learning paradigm in Table 4. Comparing the results of the last two rows, we can observe that our model maintains the inference speed while effectively improving the restoration quality after embedding this strategy.

4.5.4. Effect of different knowledge priors

To assess the impact of knowledge priors on our distillation strategy of SmoRestor, we compare several sources, including SAM (Kirillov et al., 2023), MedSAM (Ma et al., 2024), and MobileSAM (Zhang et al., 2023). As shown in Table 5, using no semantic guidance yields the lowest performance, confirming the importance of segmentation priors. SAM achieves better results, benefiting from strong generalization and precise mask quality. MedSAM also improves performance significantly, especially in medical settings, while MobileSAM offers a better trade-off between accuracy and computational efficiency. Overall, SmoRestor embedded in different SAM-based models shows similar performance for desmoking. Finally, we adopt SAM, which performs slightly better, as the default large vision model that tends to have better generalization.

4.5.5. Number of regional transformer block

We further investigate the impact of varying the number of frequency-decomposed regional transformer blocks in our framework.

Table 6
Ablations on the number of Fourier-prior-embedded regional transformer blocks.

# Transformer blocks	PSNR ↑	SSIM ↑	LPIPS ↓
2 blocks	22.9121	0.8084	0.1701
3 blocks	23.6725	0.8391	0.1522
4 blocks	**24.6192**	**0.8834**	**0.1496**
5 blocks	24.1184	0.8710	0.1547

Table 7
Ablations on the optical flow-based reconstruction loss $\mathcal{L}_{rec}$ in SmoRestor.

Loss function	PSNR ↑	SSIM ↑	LPIPS ↓
w/o $\mathcal{L}_{rec}$	22.4321	0.7724	0.1835
with $\mathcal{L}_{rec}$	**24.6192**	**0.8834**	**0.1496**

Table 8
Comparison for our frequency-decomposed modeling with related modules.

Modules	PSNR ↑	SSIM ↑	LPIPS ↓
ResNet Blocks (He et al., 2016)	20.1653	0.7316	0.2398
Restormer-style attention (Zamir et al., 2022)	22.7021	0.8001	0.1756
Frequency-decomposed attention	**24.6192**	**0.8834**	**0.1496**

As shown in Table 6, our model achieves better performance with 4 blocks. Using only two or three blocks leads to a noticeable drop in restoration quality, likely due to insufficient representation capacity. Interestingly, increasing the depth to five blocks does not bring further gains and slightly degrades performance, which we attribute to possible overfitting and increased optimization difficulty. The result indicates that a depth of four blocks can offer a good balance between accuracy and complexity.

4.5.6. Effect of alignment-aware reconstruction loss

To tackle the issue of misalignment in unpaired input and ground truth images, we introduce an optical-flow-based alignment mechanism into our reconstruction loss $\mathcal{L}_{rec}$. Specifically, we use PWC-Net (Sun et al., 2018) to estimate the optical flow between the restored image and the ground truth, and warp the GT to align with the prediction before computing their distance. This alignment ensures that the loss calculation focuses on perceptual differences rather than spatial mismatches. As shown in Table 7, removing $\mathcal{L}_{rec}$ leads to a significant performance drop, with PSNR dropping from 24.6192 to 22.4321 and SSIM from 0.8834 to 0.7724. These results confirm that without flow-based alignment, the restoration model suffers from noticeable blurriness due to pixel misalignment. Our alignment-aware loss effectively mitigates this issue and enhances visual quality.

4.5.7. Effect of frequency-decomposed attention

To evaluate the effectiveness of our proposed frequency-decomposed attention, we perform an ablation study by replacing the original module with other alternatives: ResNet blocks (He et al., 2016) and the attention operation used in Restormer (Zamir et al., 2022). As shown in Table 8, both replacements lead to a noticeable performance drop, demonstrating the superiority of our frequency decomposition in modeling complex spatial-frequency interactions. The Restormer-style attention performs better than vanilla ResNet blocks, but still falls short of the proposed mechanism, indicating that frequency-decomposed design provides an effective representation capability in challenging surgical degradation scenarios.

4.6. Limitations

While our framework achieves strong performance on both DesmokeData and private datasets, we acknowledge two main limitations. First, surgical smoke remains highly challenging due to its heterogeneous and dynamic nature. In particular, our failure cases (see Fig. 8) show that when the smoke is extremely dense and fully obscures anatomical structures, the restoration becomes unreliable and residual artifacts may persist. Second, although the proposed design improves efficiency compared to conventional transformer-based approaches, real-time deployment in clinical environments still requires further optimization, as computational latency may limit seamless integration into surgical video systems.

For future work, we plan to validate the framework on larger multi-center datasets with diverse imaging devices and surgical practices to better establish generalizability. We also aim to extend the method to handle other clinically relevant degradations in a unified manner and to integrate it as a preprocessing module for downstream tasks such as surgical workflow recognition, bleeding detection, and anatomical localization. Exploring these extensions in real surgical settings represents a promising avenue for future research.

5. Discussion

Requirements. Surgical desmoking remains a distinctive and technically demanding restoration task due to the complex visual artifacts introduced by cauterization. Unlike degradations in natural scenes, surgical smoke is highly dynamic, non-uniform, and interacts intricately with tissue texture and lighting conditions. These characteristics make conventional enhancement or haze-removal strategies ineffective. A dedicated desmoking framework must therefore account for both the temporal continuity of smoke diffusion and the fine structural preservation of anatomical regions.

Difficulties. The primary challenge lies in accurately separating smoke layers from underlying tissue without disrupting anatomical fidelity. The semi-transparent and spatially variant nature of smoke leads to ambiguous depth cues and irregular illumination, which hinder both local texture recovery and global contrast consistency. Furthermore, the rapidly changing camera motion and tool-induced reflections exacerbate the instability of smoke estimation. These factors collectively create an underconstrained restoration problem that demands models capable of discriminating degradative and anatomical signals in a temporally coherent manner.

Prospects. Future work should explore degradation modeling grounded in the physics of smoke formation and tissue reflectance, rather than relying solely on data-driven priors. Leveraging temporal cues and motion consistency could enhance stability across frames. Moreover, integrating flow-guided or physically inspired constraints may improve the disentanglement between smoke and true structures. Such direction-specific developments will be instrumental in advancing desmoking research toward more robust and interpretable restoration frameworks.

6. Conclusion

This paper presents a dedicated framework for surgical image desmoking, addressing the core challenges of surgical degradation modeling, computational efficiency, and lightweight deployment. We concentrate on the complex restoration problem of smoke-induced surgical degradation and propose an efficient high-order frequency-decomposed transformer that effectively disentangles anatomical content from dense smoke artifacts. Additionally, we embed a KV-cache mechanism to enhance inference efficiency and a novel knowledge transfer paradigm that distills large vision model capabilities into compact networks. Our SmoRestor provides a robust and practical solution for real-time surgical desmoking, integrating theoretical insights, semantic priors, and efficient learning strategies to advance surgical image restoration.

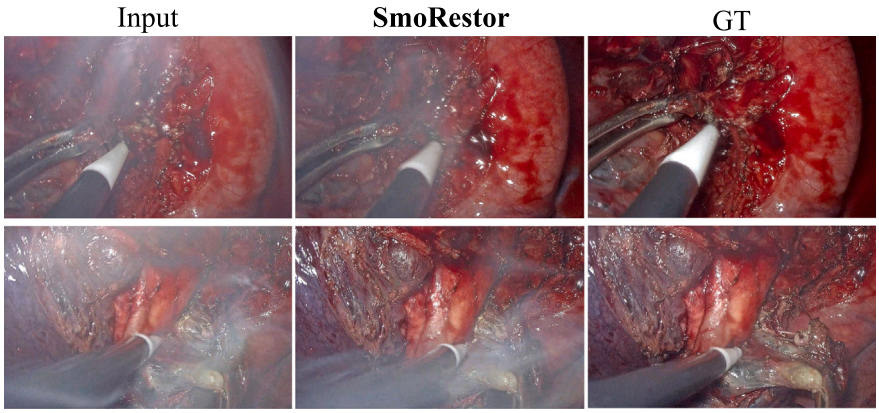

Fig. 8. Failure cases of SmoRestor with dense smoke in our in-house dataset.

CRediT authorship contribution statement

Jiaao Li: Writing – original draft, Investigation. **Diandian Guo:** Software, Methodology. **Youyu Wang:** Data curation. **Yanhui Wan:** Validation, Supervision, Funding acquisition. **Long Ma:** Writing – review & editing, Methodology. **Jialun Pei:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

We use a publicly available dataset and our in-house datasets for evaluation.

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